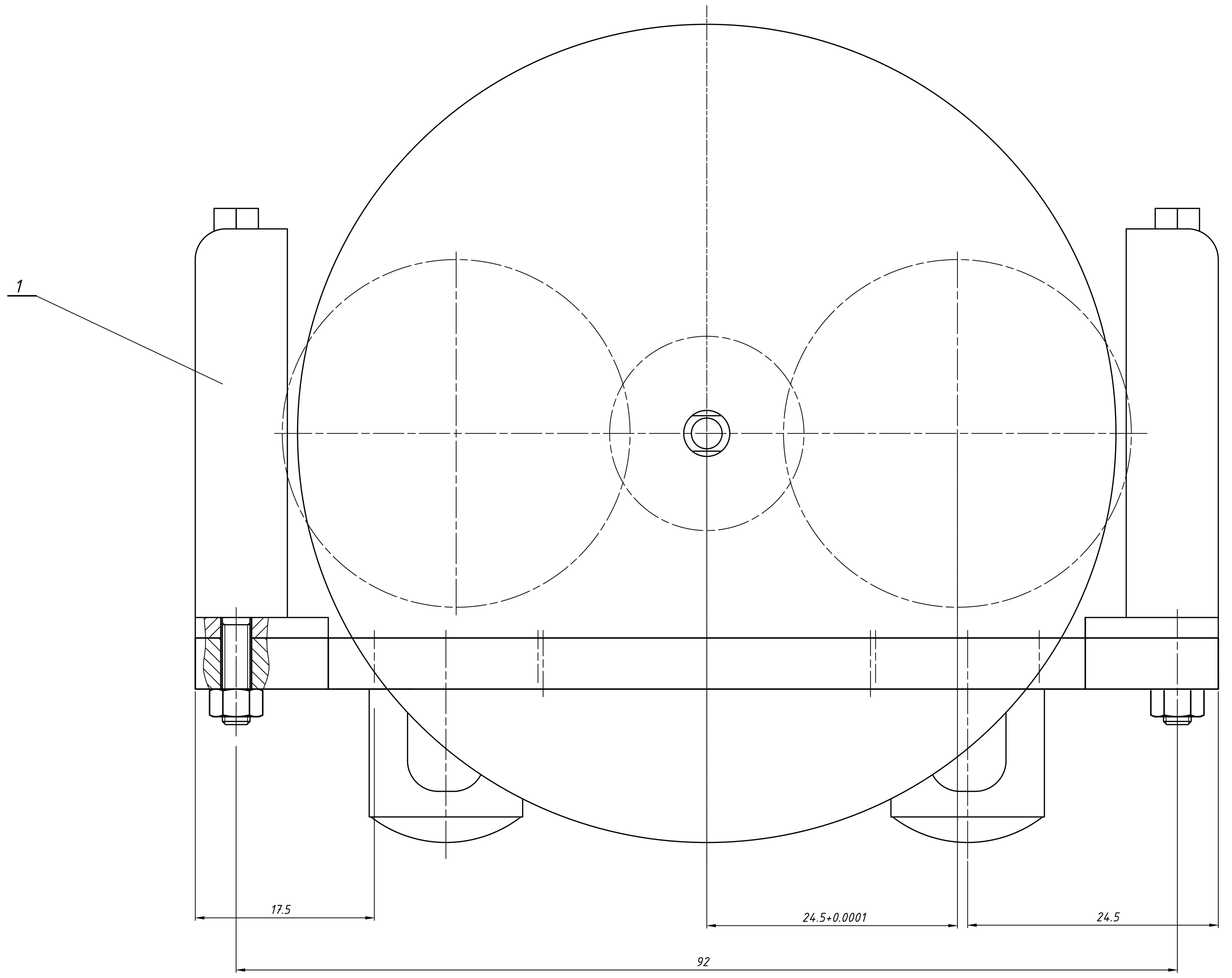

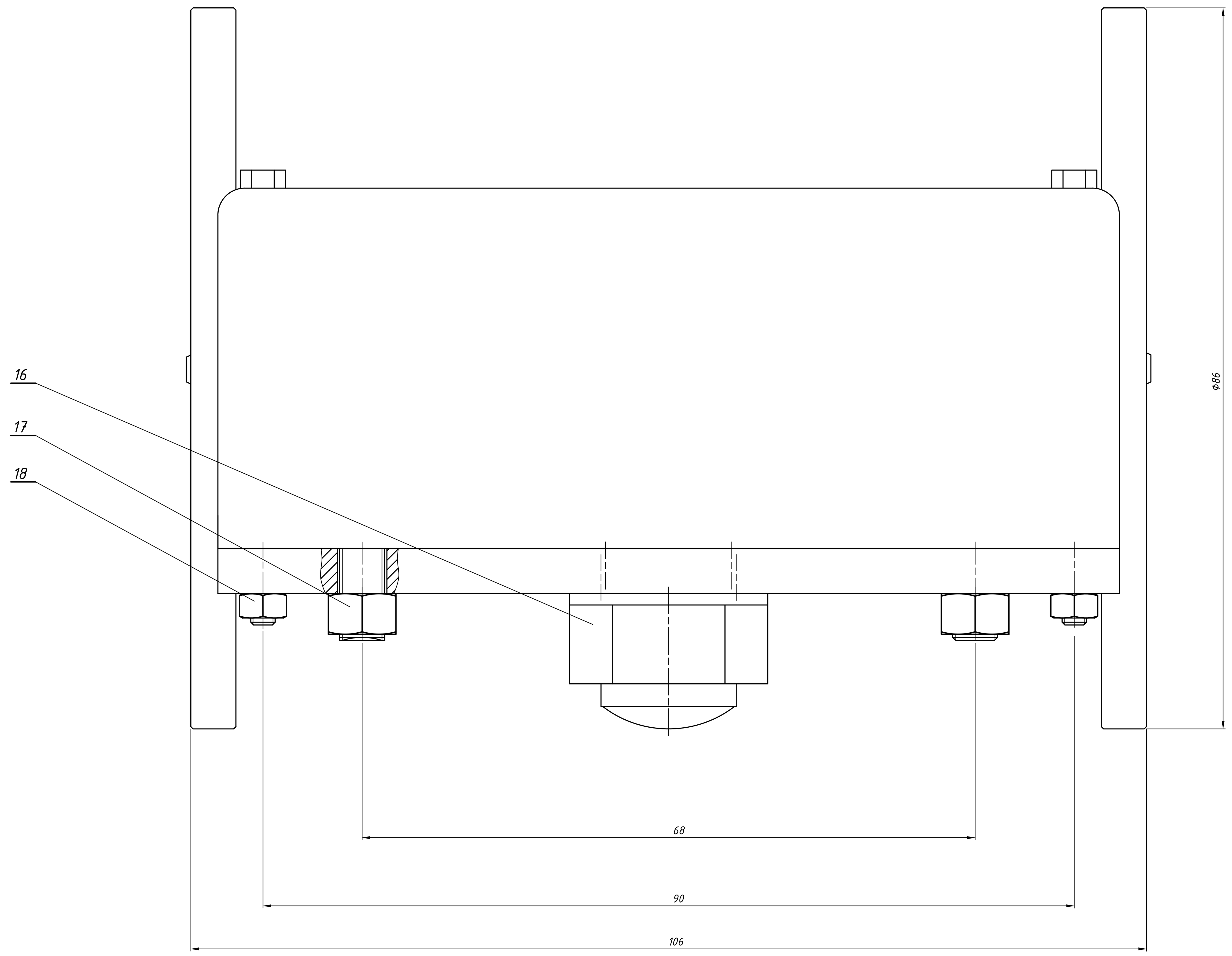

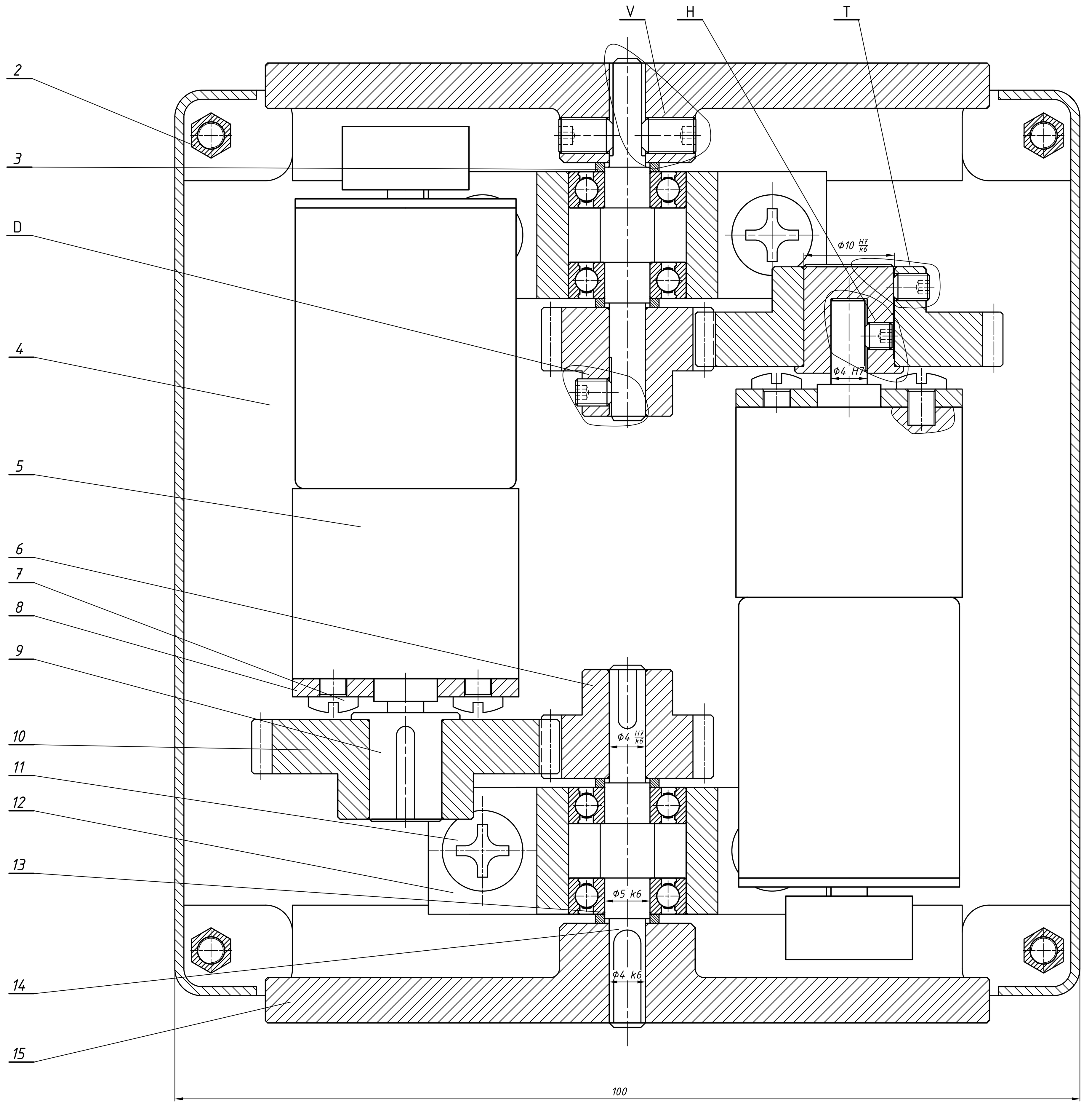

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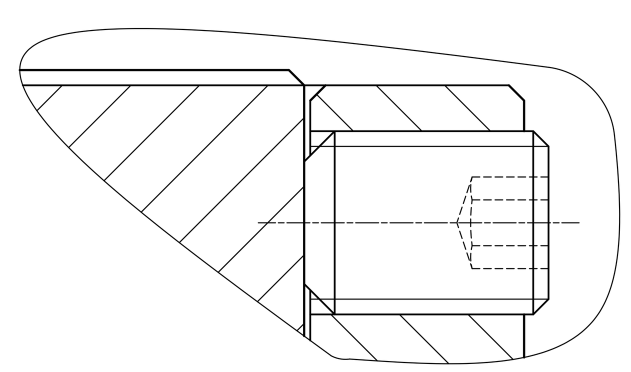

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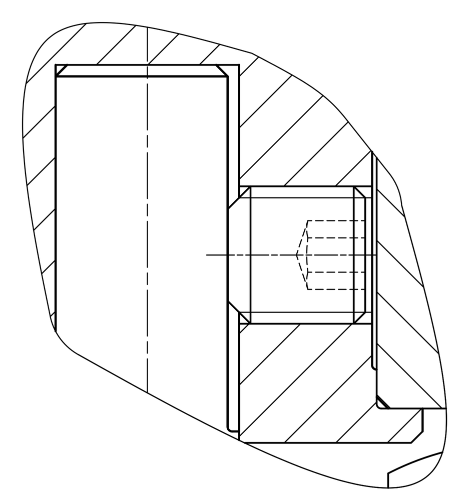

V
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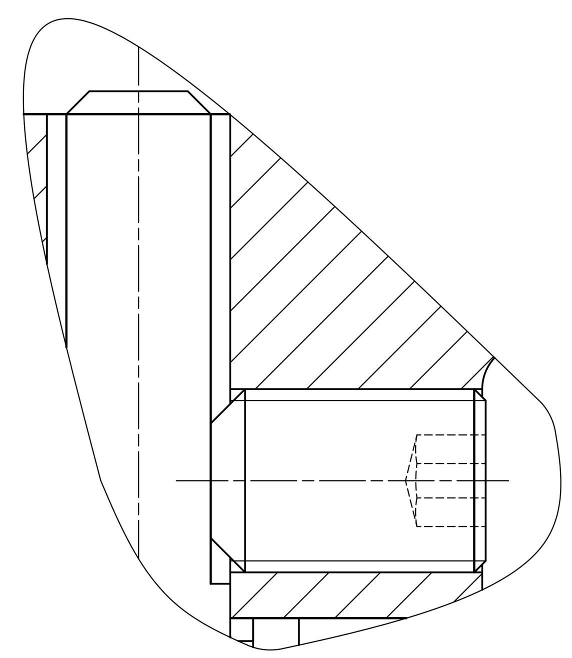

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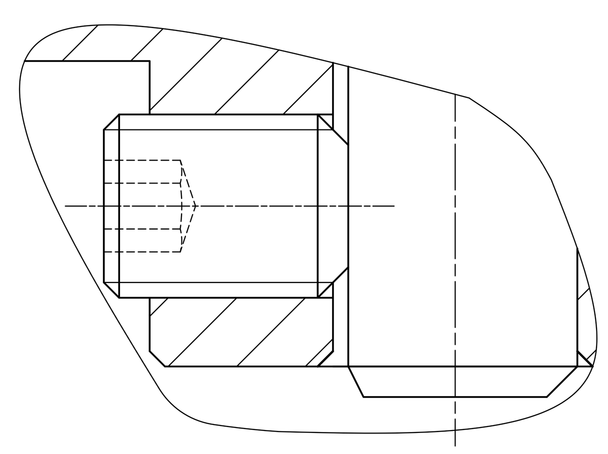

TECHNICAL TABLE OF THE GEARBOX

Shaft / Parameter	Motor	I	
Transmission ratio	0.5		
Rotation speed (rpm)	250	500	
Power (W)	9.63	8.6	
Moment M (N.mm)	368	164	

TECHNICAL REQUIREMENTS

1. The wheel requires 2 screws to ensure the balance
2. Each gear need a screw instead of a key to be mounted
3. The installation need to ensure the shaft coaxial problem.

No.	Symbol	Name	Q.ty	Material	Note
D	SSHHD-ST-M3-4	Set Screw M3	4	Stainless Steel	SUNCO
V	SSHHD-ST-M4-6	Set Screw M4	2	Stainless Steel	SUNCO
H	SSHHD-ST-M3-3	Set Screw M3	2	Stainless Steel	SUNCO
T	SSHHD-ST-M3-4	Set Screw M3	2	Stainless Steel	SUNCO
18	HNT1-A-600-M3	Hex nut M3	4	Steel S45C	SUNCO
17	HNT2-ST3B-M5	Hex nut M5	4	Steel S45C	SUNCO
16	HS000073	Passive wheel	1	Plastic	
15	ME08	Wheel	2	Pure Plastic	
14	ME07	Shaft	2	Steel C45	
13	619/5	Deep groove ball bearing	4	Steel 100CR6	SKF
12	SK13	Shaft housing	2	steel	MISUMI
11	B040515	Truss head machine screw	4	Low carbon steel (SWCH)	
10	ME06	Driving gear	2	steel C45	
9	ME05	Coupling	2	steel	
8	DIN 7985	Combination screw	4	brass	
7	HS000524	DC geared motor mounting bracket	2	aluminum	
6	ME04	Driven gear	2	steel C45	
5	GA25	DC gearbox Motor encoder	2	steel	
4	ME03	Base	1	PLA	
3	ME02	Machine element 02	4	GX15-32	
2	X6ZC	Cooper bar	4	cooper	
1	ME01	Robot Cover	2	PLA	

THESIS

DESIGN SOCCER ROBOT

	Name	Signature	Date		Quantity	Weight	Ratio
Designed by	Bui Viet Thang			Soccer robot	01		4 : 1
	Phan Tien Minh						
Instructed by	Pham Cong Bang				Size: A0	Sheet: 1	
Approved by					HCMC University of Technology Faculty of Mechanical Engineering		